

PEYTON RAMSEY

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EDUCATION

University of New Hampshire	Durham, NH
MS in Business Analytics (MSBA), GPA: 3.63	August 2025 – May 2026
BS in Business Administration: Focus in Information Systems & Business Analytics	August 2021 – May 2025
<i>NCAA Division I Cross Country, Indoor Track & Outdoor Track; Dean's List; America East Commissioner's Honor Roll</i>	

EXPERIENCE

Graduate Research Assistant, Data Science & AI	August 2025 – May 2026
<i>University of New Hampshire</i>	<i>Durham, NH</i>

- Designed hands-on labs and course content for AI Tools and Apps for Business covering ML deployment, LLM workflows, Git, and Docker, giving MSBA students direct exposure to production-ready tooling.

Data Engineer Intern View TPA Compliance Report	June 2025 – August 2025
<i>Nassau Financial Group (Insurance / Financial Services)</i>	<i>Hartford, CT</i>

- Built and deployed a self-serve R Shiny compliance review platform that prioritized higher-risk cases across 120K+ insurance policies from 7 AWS data streams, reducing manual review time by about 60%.
- Engineered ETL pipelines and a configurable weighted scoring engine across 8 insurance activity types; partnered with compliance and stakeholders to deliver a production tool used by 30 analysts and presented to 50+ stakeholders.

Business Intelligence & Data Intern	June 2024 – August 2024
<i>Sprague Operating Resources (Energy / Natural Gas)</i>	<i>Portsmouth, NH</i>

- Wrote SQL to build star-schema models in Power BI from trade and pricing data, eliminating a 30+ minute manual pull from a legacy system and enabling on-demand portfolio pricing analysis for the sales team.

SELECTED DATA SCIENCE PROJECTS

Airbnb Booking Prediction & Market Analysis View Report	March 2025 - May 2026
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- Scaled Airbnb booking-conversion modeling from 12,947 SF listings to 374K U.S. listings and a 93K-row Paris panel, achieving 0.917 ROC-AUC, 0.8668 Kaggle AUC, and 0.966 holdout AUC with leakage-safe tabular and NLP features.
- Applied TPC causal structure learning and Double ML with random forest residualization to validate feature logic, removing downstream leakage such as reviews per month and deprioritizing non-causal signals before training.
- Identified a ~6x booking-rate gap between commercial and casual hosts that price and location could not close, pointing to host operating model as a structural driver of performance.

Financial News Retrieval and Sentiment Analysis (RAG) Live Demo	January 2026 - Present
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- Built a RAG system for financial news using dense retrieval and FinBERT sentiment analysis to answer questions on sectors, tickers, and macro themes, achieving 92% retrieval precision and 90% answer accuracy across 30 test queries.
- Designed a 5-stage pipeline with AB testing to filter 500+ raw articles into a 50-document knowledge base and tuned retrieval settings to reach a 2.5 to 4.5 second end-to-end latency.
- Automated weekly article refresh with GitHub Actions and an AI agent, then deployed with Docker on Hugging Face Spaces.

Macroeconomic Anomaly Detection Live Demo	January 2026 - March 2026
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- Built a multivariate LSTM autoencoder on 7 FRED indicators to detect macroeconomic regime shifts from reconstruction error, without labeled anomalies or crisis dates in training.
- Calibrated a 0.0203 threshold from a pre-2007 baseline, flagged all 3 target stress periods in historical evaluation, and deployed on Hugging Face Spaces with MLflow tracking and a Llama-3.1-8B explanation layer.

TECHNICAL SKILLS

Programming: Python (Pandas, NumPy, Polars, PyTorch, scikit-learn), SQL, R (tidyverse, dplyr, ggplot2)

Machine Learning & Statistics: supervised & unsupervised learning, classification & regression, feature engineering, model evaluation & cross-validation, tree-based & boosting methods (Random Forests, LightGBM, CatBoost), SHAP, NLP, deep learning (LSTM autoencoders), clustering (k-means, DBSCAN), time series analysis (ARIMA, VECM), A/B testing

MLOps & Tools: FastAPI / Flask, MLflow, Docker, Git/GitHub, Hugging Face, R Shiny, Power BI, AWS